SEGi UNIVERSITY

FACULTY OF ENGINEERING, BUILT ENVIRONMENT AND INFORMATION TECHNOLOGY

BIT2233/BTL2233/BCL2233 COMPUTER ARCHITECTURE

CONTINUOUS ASSESSMENT 30%: Assignment 2

Computer Architecture Analysis Based on Professional or Industry Context

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I hereby declare that this lab assessment submission is my own independent work and does not contain

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PART A: Professional Context Identification

The professional context for this report is the **Retail Pharmacy Sector**, specifically focusing on the operational environment of a community pharmacy. In this high-stakes industry, the accuracy of medication dispensing, inventory management, and patient safety are paramount. The chosen computing system for analysis is the **Xilnex Point-of-Sale (POS) and Retail Management System**.

Xilnex is a cloud-native, omnichannel solution designed to streamline pharmacy operations. It integrates essential front-end retail functions with back-end pharmacy management, including unified inventory tracking across multiple branches, Customer Relationship Management (CRM) for patient loyalty, and secure payment processing. Its "offline-first" architecture ensures that pharmacists can continue to serve patients even during internet disruptions, synchronizing data to the cloud once connectivity is restored (Xilnex, 2022). This system is critical for maintaining real-time stock accuracy, which is vital for managing drug expiries and controlled substances.

PART B: Architecture Component Analysis

2.1 CPU Role and Functions

The Central Processing Unit (CPU) within the Xilnex POS terminal serves as the primary execution engine for all local operations. In a pharmacy context, the CPU performs complex logic calculations such as FEFO (First-Expiry, First-Out) logic to select medication batches and handles AES-256 encryption for data security (Sarcouncil, 2025). It manages the fetch-decode-execute cycle for the POS software, ensuring smooth UI updates and background cloud synchronization.

2.2 Memory and Storage Architecture

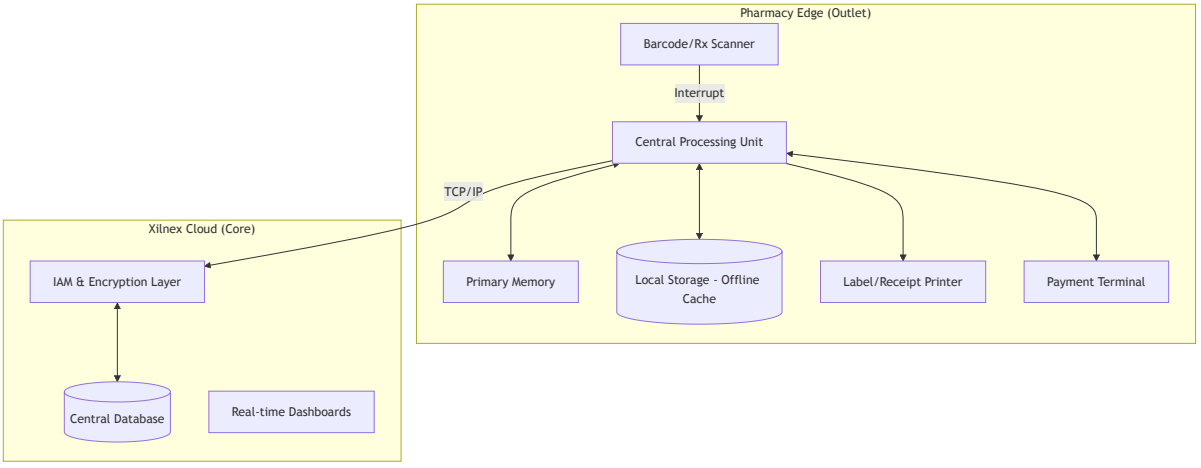
The system employs a multi-tiered memory hierarchy:

- **Primary Memory (RAM):** 8GB to 16GB for active session data and latency reduction.
- **Secondary Storage (Local SSD):** Stores the offline database cache for resilience (arXiv, 2026).
- **Cloud-Based Storage:** The central repository for global inventory and historical patient data.

2.3 Input and Output (I/O) Devices

Specialized I/O peripherals include 2D Barcode Scanners for drug authentication, Thermal Label Printers for prescription instructions, and integrated payment terminals (EFT) for secure transactions.

2.4 Data Flow and Architecture Diagram



PART C: I/O Organisation & Integration Trade-Offs

3.1 Polling vs Interrupt Mechanisms

For input devices like the barcode scanner, an **Interrupt-driven I/O** mechanism is used to ensure zero-latency feedback. In contrast, the **Cloud Synchronization** process employs **Polling** or Webhooks to check network status periodically, which is safer for managing asynchronous network I/O (ResearchGate, 2024).

3.2 Real-Time vs Batch Processing

Real-time processing is essential for immediate stock deduction during sales, while batch processing is reserved for non-critical tasks like End-of-Day reconciliation to reduce CPU load during peak hours (arXiv, 2026).

3.3 Security vs Usability Trade-Offs

Implementing Multi-Factor Authentication (MFA) enhances security but can slow down staff. The pharmacy sector justifies this "friction" as the cost of a data breach outweighs the minor inconvenience of login delays (Chavan & Bhoite, 2024).

PART D: Architecture Recommendation & Justification

I recommend implementing a **Hybrid Edge AI Module** and **Biometric SSO**.

Justification: Local AI validation ensures safety checks function during outages, while biometrics provide high security with instant access, improving both reliability and efficiency (Sarcouncil, 2025).

LIST OF REFERENCES

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